

Picture Postcard — Game Design Document

Cognitive Training Suite for Older Adults (60-80) • Memory Domain • Game 1 of 6

Version: 1.0 • Status: For implementation • Platform: iOS / Android tablet-first, phone-supported

1. Purpose and Design Intent

Picture Postcard is a **change-detection and location-binding memory trainer**. The player studies a rich illustrated scene, the scene is briefly withdrawn, and a modified version returns. The player must detect, localise, or reconstruct what changed.

The design targets three trainable capacities:

Capacity	How the game loads it
Visual working memory	Limited encoding window forces prioritised, strategic encoding rather than exhaustive scanning
Object-location binding	Player must recall <i>what</i> was <i>where</i> , not just <i>what</i>
Episodic recall under delay & interference	Retention intervals, masking, and within/across-session probes

Difficulty philosophy. The ramp is deliberately steep. The training-relevant zone is where the player fails roughly one trial in five — comfort produces no adaptation. Steepness is delivered through *task demand*, never through reaction-time pressure, small targets, or ambiguous art. The player should always feel the task is fair and their own effort is the variable.

Non-negotiable: the player must never experience a hard failure state. Difficulty falls to meet them; scaffolding rises to meet them. Progress is never lost.

2. Player Profile and Accessibility Baseline

Designed for ages 60–80, including users with presbyopia, mild tremor, reduced contrast sensitivity, and variable touchscreen fluency.

- **Touch targets:** minimum 64 dp visual, 96 dp hit area. Hit radius 1.5× the visual bound.
 - **Type:** body ≥ 20 sp, instructions ≥ 26 sp, no light font weights. System font-scaling respected.
 - **Contrast:** $\geq 4.5:1$ for all text and interactive edges. Never encode meaning in hue alone (red/green deficiency is common at this age).
 - **Interaction:** single tap only through Tier 6. Drag introduced at Tier 6 with generous snap (48 dp magnet radius) and tap-select-then-tap-place as an equivalent alternative.
 - **Motion:** no flashing above 3 Hz. Ambient scene motion is a difficulty parameter and must be disableable in Accessibility settings without blocking progression (substitute an equivalent load axis — see §5.3).
 - **Audio:** all feedback dual-coded visually. Nothing is audio-only.
 - **Session control:** pause is always available and never penalised. The game auto-saves after every round.
 - **Tone:** no cartoon failure, no "you lost", no red X on a face. Errors are neutral and informative.
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3. Core Loop

Round structure (one trial):

1. **Ready** — "Look carefully" card, 1.5 s. Shows the round number and the probe type to expect.
2. **Encoding** — scene displayed for `encodeMs`. A slim depleting bar shows time remaining (progress, not threat — no red, no ticking sound).
3. **Retention** — scene replaced by a neutral pattern mask for `delayMs`. From Tier 6, an interference micro-task occupies this window.
4. **Probe** — one of five response modes (§4.2). Player responds by tapping.
5. **Feedback** — immediate, ≤ 2 s. Correct: the target region blooms with a soft ring, score ticks up. Incorrect: the correct region is revealed and the original scene is re-shown for 1.2 s with the change annotated. *Error trials always end with a corrective re-exposure — this is the learning event, and it must never be skippable by accident.*

Level structure: 10 rounds per level. Session = 1-2 levels, 6-9 minutes.

Level completion:

Accuracy	Outcome
$\geq 90\%$	3 stars, advance
70-89%	2 stars, advance
50-69%	1 star, advance
$< 50\%$	Level repeats with regenerated content at -1 difficulty step; no stars lost, no lives lost

Advancement is generous by design; **stars, not gates, carry the achievement signal.**
Blocking an older adult behind a wall produces dropout, not learning.

4. Difficulty Model

4.1 The seven axes

Every level is fully described by a parameter vector. This is the contract for content generation.

#	Axis	Symbol	Range	Notes
1	Scene set size	<code>objects</code>	5 → 18	Count of <i>salient, manipulable</i> objects, excluding background
2	Encoding time	<code>encodeMs</code>	8000 → 3000	Per-object budget falls from ~1.6 s to ~0.17 s
3	Retention delay	<code>delayMs</code>	500 → 10000	Masked from Tier 4
4	Simultaneous changes	<code>changes</code>	1 → 4	Player must find all before the round resolves
5	Change-type class	<code>changeTypes []</code>	Class 1 → 10	See table below
6	Distractor similarity	<code>lureLevel</code>	0 → 3	Near-identical objects placed near the target
7	Probe mode	<code>probeMode</code>	M1 → M5	See §4.2

Change-type difficulty classes (ordered by empirical detection difficulty):

Class	Change type	First appears
1	Object removal, central & large	L1
2	Object addition	L4
3	Object translocation, large displacement	L11
4	Colour change	L15
5	Object substitution (same category — different dog, different bicycle)	L21
6	Scale change	L33
7	Orientation / mirror flip	L41
8	Peripheral or small-object change	L55
9	Gradual change (object morphs slowly during encoding, no transient)	L71
10	Relational change (two objects swap positions; scene inventory identical)	L81

Class 10 is the hardest because no single object is "wrong" — only the *relationship* is. It is the purest test of binding.

4.2 Probe modes

Mode	Task	Memory demand
M1	Both scenes visible; tap what differs	Recognition, comparison-supported

Mode	Task	Memory demand
M2	Only modified scene visible; tap where the missing object was	Spatial recall
M3	Neither scene visible; cued multiple choice ("What colour was the bicycle?" — 4 image options)	Cued recall
M4	Modified scene visible with an object tray; drag/tap the correct object back to its correct position	Free recall + source binding
M5	Delayed probe about a scene from 2–3 rounds earlier, or from the previous session	Episodic consolidation

M5 is the highest-value mechanic for real memory transfer and is the game's signature. From Tier 8, every level closes with one M5 round; from Tier 10, one M5 round references the **previous day's session**, framed warmly ("Do you remember yesterday's harbour?").

4.3 Parameter curves

For any level $n \in [1, 100]$, nominal values are:

```

objects    = round( 5 + 13 * (n/100)^0.85 )
encodeMs   = round( 8000 - 5000 * (n/100)^0.70 )
delayMs    = round( 500 + 9500 * (n/100)^1.20 )
changes    = 1 + floor(n / 28)
lureLevel  = floor(n / 26)

```

These are the **nominal** values. Actual values in play are the nominal values shifted by the adaptive engine (§5).

5. The 100-Level Ladder

Ten tiers of ten levels. Within a tier, levels 1–10 interpolate linearly between the tier's entry and exit values while rotating scene themes and change types.

Tier	Levels	Objects	Encode	Delay	Changes	New change classes	Probe modes	Time
1 · Foundations	1-10	5 → 7	8.0 → 7.0 s	0.5 → 1.0 s	1	1, 2	M1	none
2 · Binding	11-20	7 → 8	7.0 → 6.2 s	1.0 → 2.0 s	1	3, 4	M1, M2	none
3 · Recall	21-30	8 → 10	6.2 → 5.6 s	2.0 → 3.0 s	1	5	M2	soft 60 s
4 · Divided load	31-40	10 → 11	5.6 → 5.1 s	3.0 → 4.2 s	2	6	M2, M3	soft 60 s
5 · Interference	41-50	11 → 12	5.1 → 4.7 s	4.2 → 5.5 s	2	7	M3	soft 45 s
6 · Reconstruction	51-60	12 → 13	4.7 → 4.3 s	5.5 → 6.5 s	2	8 (L55)	M3, M4	soft 45 s
7 · Clutter	61-70	13 → 15	4.3 → 3.9 s	6.5 → 7.5 s	3	—	M4	soft 45 s
8 · Consolidation	71-80	15 → 16	3.9 → 3.6 s	7.5 → 8.3 s	3	9	M4, M5	soft 40 s
9 · Relational	81-90	16 → 17	3.6 → 3.3 s	8.3 → 9.2 s	3	10	M3, M4, M5	soft 40 s
10 · Mastery	91- 100	17 → 18	3.3 → 3.0 s	9.2 → 10.0 s	4	all mixed	all, unannounced	soft 40 s

Notes for the implementer:

- *Soft timer* means the round resolves when time expires but is scored as an omission, not a failure; no countdown audio, and the last 10 s are not visually escalated. Its purpose is session pacing, not pressure.
 - **Interference micro-task** (Tiers 5+): during the retention interval the player answers a trivially easy unrelated question ("tap the larger circle"). This blocks rehearsal and is the single most effective difficulty lever available without shortening encoding further.
 - The per-object encoding budget deliberately collapses from ~1.6 s to ~0.17 s. **This is intentional.** Beyond Tier 5 exhaustive encoding is impossible; the trainable skill becomes prioritised allocation and gist-plus-detail chunking. The game should teach this explicitly once, at L41, in a 20-second tip card.
 - **Ambient motion** (Tier 6+) — gentle scene animation (drifting clouds, a flag, water shimmer) never involves the target object. If disabled for accessibility, add +1 to `objects` instead.
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6. Adaptive Difficulty Engine

6.1 Within-level staircase

A **transformed up-down staircase** on a continuous Difficulty Index (DI), targeting ~79% accuracy — the empirically favourable training zone.

```
3 consecutive correct → DI += 1 step
1 incorrect           → DI -= 1 step   (-2 steps if DI is above nominal)
```

One DI step $\approx 8\%$ of the delta between adjacent levels. The staircase adjusts, in priority order: `delayMs` $\rightarrow$ `encodeMs` $\rightarrow$ `lureLevel` $\rightarrow$ `objects`. Object count changes last, because changing it mid-level is visually conspicuous and feels like the game "noticing" the player struggling.

6.2 Bounded adaptation

DI is clamped to nominal ± 2 levels. The ladder, not the staircase, owns long-run progression. Without this clamp a struggling player is silently parked at Level 12 difficulty while the UI says Level 60 — which destroys both the felt sense of progress and the research validity of level number as a covariate.

6.3 Cross-session carry

At session start, DI initialises at the *rolling mean of the last 20 trials*, minus one step. Older adults show a reliable warm-up cost; opening a session at yesterday's ceiling reads as regression and is demoralising. The first two rounds of any session are always warm-up rounds at -2 steps and are flagged in telemetry so they can be excluded from outcome analysis.

6.4 Frustration guard

Three consecutive incorrect trials trigger a **confidence round**: one trial at nominal -4 , then the staircase resumes from nominal -2 . This fires at most twice per level. It is not announced.

7. Scaffolding — "Subtle Helpers"

Assistance escalates automatically and silently. The player should experience it as the game being gentle, not as being helped.

Tier	Trigger	Assistance
0	—	None
1	6 s with no response	Non-target regions desaturate by 8% — imperceptible as an event, but it narrows the effective search field
2	12 s, or 1 wrong tap	Target quadrant gains a soft warm vignette
3	2 wrong taps	Target object pulses at 25% opacity, twice, 0.4 s each
4	3 wrong taps	0.75 s "peek" at the original scene, then the answer is revealed with annotation

Explicit Hint button: 3 per level, non-refillable within the level, jumps straight to Tier 3. It costs score, never progress. Unused hints convert to bonus points at level end — this rewards restraint without punishing use.

Design rule: scaffolding never appears before 6 s. Premature help is the most common way cognitive-training apps destroy their own effect size — the effortful retrieval attempt is the training.

8. Scoring and Feedback

```
roundScore = 100 × changesFound
              + speedBonus      (0-40, Tier 3+ only, from the soft-timer
remainder)
              + streakMultiplier (×1.0 / ×1.2 / ×1.5 at streaks of 3 / 5 / 8)
```

- 30 per hint used
- 15 per scaffolding tier auto-triggered above Tier 1

- Score is **never negative** and the floor for a completed round is 10 points.
- No global leaderboards. Comparison is against the player's own 7-day rolling average only.
- Session-end summary is short, warm, and states one concrete thing: *"Your best today: the harbour scene with 12 objects. That's two more than last week."*
- A weekly "Brain Map" shows progress across the six-game suite. It presents *change over time*, never a percentile against other users, and never a cognitive-age or clinical-sounding figure.

9. Content Requirements

Scene library: 30 base scenes across 10 familiar themes — park, seaside, kitchen, village square, garden, market, festival, railway platform, café, harbour. Each theme should feel warm, populated, and everyday. Avoid: children as focal subjects, brand-like signage, dense text, anything requiring cultural insider knowledge.

Localisation note: for an India-based deployment, at least one third of scenes should be regionally recognisable (local market, temple street, hill station, tea stall). Scene familiarity affects encoding efficiency and is a genuine confound if the whole library is Western.

Per-scene authoring requirements:

- 20+ authored object slots per scene, each with: `(id)`, `(category)`, `(bbox)`, `(salience)` (1-3), `(centrality)` (1-3)
- Each object needs 3+ variants for substitution: colour variants, a same-category alternate, a mirrored version, a 0.8× / 1.25× scaled version

- Two valid alternate positions per movable object
- Layer separation so any object can be removed cleanly without background artefacts
- A "near-lure" partner for each object (a visually similar item that can be placed nearby at $\text{lureLevel} \geq 2$)

Thirty scenes at this authoring depth yield **well over 100,000 distinct trials**, which is sufficient for a multi-month intervention with no repetition and, critically, prevents item-specific learning from masquerading as cognitive gain.

Anti-repetition rule: no scene repeats within a session; no scene-change pairing repeats within 30 days.

10. Data Model

Level definition

json

```

{
  "level": 47,
  "tier": 5,
  "rounds": 10,
  "nominal": {
    "objects": 12,
    "encodeMs": 4900,
    "delayMs": 5100,
    "changes": 2,
    "lureLevel": 1,
    "interference": true,
    "ambientMotion": false,
    "softTimerMs": 45000
  },
  "changeTypeWeights": { "1":0.05, "2":0.05, "3":0.15, "4":0.20, "5":0.25, "6":0.30 },
  "probeModeWeights": { "M2":0.2, "M3":0.8 },
  "themePool": ["seaside", "market", "garden", "cafe"],
  "adaptiveClamp": { "minLevelEquivalent": 45, "maxLevelEquivalent": 49 }
}

```

Per-trial telemetry (required for research use)

(trialId), (sessionId), (participantId), (timestamp), (level), (effectiveDI), (sceneId),
 (objects), (encodeMs), (delayMs), (changeType), (probeMode), (lureLevel), (correct),
 (responseLatencyMs), (tapCoordinates[]), (errorDistancePx) (distance from tap to true
 target — a graded measure of partial spatial memory, far more informative than binary
 accuracy), (scaffoldTierReached), (hintsUsed), (isWarmup), (interruptions).

Derived outcome measures: threshold (encodeMs) at 79% accuracy, threshold (delayMs)
 at 79% accuracy, mean (errorDistancePx), M5 delayed-recall accuracy, and set-size

capacity estimate k . These four are the game's primary endpoints and should be reportable per participant per week.

11. Prescription and Scheduling

- **Dose:** 15-20 minutes/day, 5 days/week — two Picture Postcard sessions of ~7 minutes, or one paired with another domain game.
 - **Spacing:** the two daily sessions should be ≥ 3 hours apart where possible; spaced practice outperforms massed practice for consolidation.
 - **Reminders:** one per day, at a user-chosen time, dismissible forever in two taps. No streak-shaming, no loss framing, no push notification that implies decline. Missing days must never reset progress.
 - **Return-after-absence:** after 7+ days away, insert one calibration level at DI -3 before resuming. Never comment on the absence.
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12. Screen Inventory

Screen	Contents
Home	Continue (large, primary), level ring, streak-free "days played this week", settings
Level intro	Level number, theme preview, one-line goal, Start
Encoding	Full-bleed scene, depleting bar, pause
Retention	Neutral mask, or interference micro-task

Screen	Contents
Probe	Scene + probe prompt + response affordance (tap zones / MCQ tray / object tray)
Feedback	Annotated reveal, score delta, Next
Level summary	Stars, accuracy, one personalised sentence, Continue / Done
Weekly Brain Map	Trend lines across the six games, plain language only
Settings	Text size, motion, sound, colour-safe palette, reminder time, gentle mode (4-up/1-down staircase → ~84% accuracy)

13. Open Questions for Pilot

1. Are the Tier 8–10 encoding times (3.0–3.9 s at 15–18 objects) achievable, or does accuracy floor out? Pilot with $n \approx 15$ before committing art budget to high-tier scenes.
2. Does the interference micro-task feel like an unfair interruption? Test both a passive mask and the active task at Tier 5.
3. Is M4 (drag-back reconstruction) motorically viable for users with tremor, or should the tap-select-tap-place alternative become the default?
4. Optimal position of the M5 cross-session probe — session open (cold recall, harder, better measure) or session close (warmer, better retention)?

14. Build Note

This document is intended to be sufficient for implementation. To generate the full ladder, apply §4.3 curves and §5 tier table to produce 100 level-definition JSON objects in the §10 schema, then validate that: (a) each tier's entry and exit values match the table, (b) no change class appears before its first-appearance level, and (c) `adaptiveClamp` bounds are ± 2 levels throughout.

Picture Postcard · Memory Domain · Game 1 of 6. Companion documents to follow: Train Yard Dispatcher, Market Shopping List, Garden Keeper, Airport Baggage Sorter, Railway Junction Master, Delivery Route Planner.